```
// Q:1
// playing with number
//Test cases
//input1=7;
//input2={1,2,3,4,5,6,7};
//input3=2;
//output : 3456712
/*
#include<iostream>
#include<vector>
using namespace std;

void shiftArray(int input1, int input2[], int input3){

vector<int> temp(input1);

for(int i=0; i<input1; i++){
temp[i] = input2[(i + input3) % input1];
}

for(int i=0; i<input1; i++){
cout << temp[i] << " ";
}
}

int main(){

int input1;
cin >> input1;

int arr[100];

for(int i=0; i<input1; i++){
cin >> arr[i];
}

int input3;
cin >> input3;

shiftArray(input1, arr, input3);

return 0;
}
*/

//Q:2 Array Sort(nick checks)
//test case:
//input1 : 6;
```

```
//input2[]:{3,7,2,5,4,6};
//output:1;

/*
#include<iostream>
#include<algorithm>
using namespace std;

int sorted(int input1, int input2[]){

sort(input2, input2 + input1);

int num = input2[0];
for(int i=0; i<input1; i++){
if(input2[i] != num) return 0;
num++;
}

return 1;
}


int main(){

int input1;
cin >> input1;

int arr[100];

for(int i=0; i<input1; i++){
cin >> arr[i];
}

cout << sorted(input1, arr) << endl;

return 0;
}
*/

//Q:3 longest number occuring count
//test case:
//n = 7;
//arr[] = {1, 2, 4, 100, 101, 3, 9}
//out : 4 {1, 2, 3, 4}
/*
#include<iostream>
#include<algorithm>
using namespace std;
```

```
int longestConsicutive(int input1, int input2[]){

sort(input2, input2 + input1);

int longest=1, current=1;

for(int i=1; i<input1; i++){

if(input2[i] == input2[i-1] + 1){
 current++;
}
else if(input2[i] != input2[i-1]){
 current = 1;
}

if(current > longest){
longest = current;
}
//cout << longest << endl;
}

return longest;

}


int main(){

int input1;
cin >> input1;

int arr[100];

for(int i=0; i<input1; i++){
cin >> arr[i];
}

cout << longestConsicutive(input1, arr) << endl;

return 0;
}
*/

//Q:4 fancy occurrence
//test case:
//input1:Welcome to metti
//input2:i
```

```
//output:Welcome to mett
/*
#include<iostream>
#include<string>
using namespace std;

string fancyOcc(string input1, string input2){

string s = "";
char com = input2[0];

for(char ch : input1){
if(ch != com) s += ch;
}
return s;
}

int main(){

string input1, input2;

getline(cin,input1);
cin >> input2;

cout << fancyOcc(input1, input2) << endl;

return 0;
}
*/

//Q:5 String within string
//test case:
//input1:abab
//input2:ab
//output: yes

/*
#include<iostream>
#include<string>
#include<vector>
using namespace std;

string toLowerCase(string str){

for(char &c : str){
if(c >= 'A' && c <= 'Z') c = c + 32;
return str;
}
```

```
}

string isPermutation(string input1, string input2){

input1 = toLowerCase(input1);
input2 = toLowerCase(input2);

vector<int> arr1(26, 0);
vector<int> arr2(26, 0);

for(int i=0; i<input1.length(); i++){
char ch = input1[i];
if (ch >= 'a' && ch <= 'z') arr1[ch-'a']++;
}

for(int i=0; i<input2.length(); i++){

char ch1 = input2[i];

if (ch1 >= 'a' && ch1 <= 'z') arr2[ch1-'a']++;

if(arr1[ch1-'a'] < arr2[ch1-'a']) return "no";
}

return "yes";
}

int main(){

string input1, input2;
cin >> input1 >> input2;

cout << isPermutation(input1, input2) << endl;

return 0;
}
*/

//Q:6 Palindrome
//input1:level
//output:1
/*
#include<iostream>
using namespace std;

int isPalindrome(string input1){
```

```
int s = 0;
int e = input1.length()-1;

while(s <= e){

if(input1[s] == input1[e]){
s++;
e--;
}
else return 0;

}

return 1;

}

int main(){

string input1;
cin >> input1;

cout << isPalindrome(input1) << endl;

return 0;
}
*/

//Q:7 Max occurring character
//test cases:
//input1:abcdd
//output: d
/*
#include<iostream>
#include<string>
#include<vector>
using namespace std;

string maxOccu(string input1){

vector<char> arr(26, 0);
string s = "";
int max = 0;

for(int i=0; i<input1.length(); i++){

char ch = input1[i];
```

```
if(ch >= 'a' && ch <= 'z') arr[ch - 97]++;

//if(max < arr[ch - 97]) max = arr[ch - 97];
}

for(int i=1; i<26; i++){

if(arr[max] < arr[i]) max = i;

}

//for(int i=0; i<26; i++){

//if(arr[i] == arr[max] && i != max) return "0";
//}

return s + (char)(max + 97);

}

int main(){

string input1;
cin >> input1;

cout << maxOccu(input1) << endl;

return 0;
}*/

//Q:8 number sum
//test case:
//input1 : 7
//input2 : {7, 2, 6, 15, 54, 10, 23} min + max
//output : 56
/*
#include<iostream>
#include<limits.h>
using namespace std;

int sum(int input1, int input2[]){

int max=INT_MIN, min=INT_MAX;

for(int i=0; i<input1; i++){
if(max < input2[i]) max = input2[i];
if(min > input2[i]) min = input2[i];
}
```

```
int ans = min + max;

return ans;

}

int main(){

int input1;
cin >> input1;

int arr[100];

for(int i=0; i<input1; i++){
cin >> arr[i];
}

cout << sum(input1, arr) << endl;

return 0;
}
*/

//Q:9 Anagram
//test cases:
//input1:build
//input : dubli
//output : yes
/*
#include<iostream>
#include<string>
#include<vector>
using namespace std;

string toLowerCase(string str){

for(char &c : str){
if(c >= 'A' && c <= 'Z') c = c + 32;
return str;
}

}

string isAnagram(string input1, string input2){

int n1 = input1.length();
int n2 = input2.length();
```

```
if(n1 != n2) return "no";

vector<int> arr1(26,0);
vector<int> arr2(26,0);

input1 = toLowerCase(input1);
input2 = toLowerCase(input2);

for(int i=0; i<n1; i++){
char ch1 = input1[i];
arr1[ch1 - 97]++;
}

for(int i=0; i<n2; i++){
char ch2 = input2[i];
arr2[ch2 - 97]++;
}

for(int i=0; i<26; i++){
if(arr1[i] != arr2[i]) return "no";
}

return "yes";

}

int main(){

string input1, input2;
cin >> input1 >> input2;

cout << isAnagram(input1, input2) << endl;

return 0;
}
*/

//Q:10 coin counts
//test case:
//input1:2
//output:5
/*
#include<iostream>
using namespace std;

int getCount(int input1){
```

```
int coin = 0;

while(input1 != 0){

coin += input1 * input1;
input1--;

}

return coin;

}

int main(){

int input1;
cin >> input1;

cout << getCount(input1) << endl;

return 0;
}
*/


//Q:11 Electro Static
//test case:
//input1: {4,3,5}
//input2: PNP
//input3: 3
//output: 600
/*
#include<iostream>
using namespace std;

int electroStatic(int input1, int input2[], string input3){

int sum = 0;
for(int i=0; i<input1; i++){

char ch = input3[i];

switch(ch){

case 'P' :
        sum += input2[i];
        break;
case 'N' :
```

```
            sum -= input2[i];
            break;
}

}

return sum * 100;

}

int main(){

int input1;
cin >> input1;

int arr[100];

for(int i=0; i<input1; i++) cin >> arr[i];

string input3;
cin >> input3;

cout << electroStatic(input1, arr, input3) << endl;

return 0;
}
*/

//Q:12 reverse Array
/*
#include<iostream>
#include<vector>
using namespace std;

vector<int> reverseArray(vector<int> input1, int input2) {
   int s = 0;
   int e = input2 - 1;

   while (s < e) {
      swap(input1[s], input1[e]);
      s++;
      e--;
   }
   return input1;
}

int main() {
   int input2;
```

```
    cin >> input2;

    vector<int> arr(input2);
    for (int i = 0; i < input2; i++) cin >> arr[i];

    vector<int> reversed = reverseArray(arr,input2);

    for (int num : reversed) {
        cout << num << " ";
    }
    cout << endl;

    return 0;
}
*/


//Q:13 modular equation
//test cases:
//input1:2
//input2:10
//input3:1025
//output:1024
/*
#include<iostream>
using namespace std;

int isModule(int input1, int input2, int input3) {
    long long power = 1;

    while (power * input1 <= input3) {
        power *= input1;
    }

    return power;
}

int main() {
    int input1, input2, input3;
    cin >> input1 >> input2 >> input3;

    cout << isModule(input1, input2, input3) << endl;

    return 0;
}
*/
```